

28/10/2024

Write a program to display contents of memory location 2000 onwards ~~at~~ ^{at} display position Do onwards. Assume look-up table for 7 seg display is stored at 3000 onwards. Unpack binary format at lower nibble position. Assume all displays are common cathode type.

MOV SI, [2000H], MOV BP, 3000H

MOV AL, 80H

OUT 86H, AL

MOV DL, 0EH

CALL DISPLAY

```

MOV DL, 0D H
CALL DISPLAY
MOV DL, 0B H
CALL DISPLAY
MOV DL, 07 H
CALL DISPLAY
HLT

```

DISPLAY :

```

MOV AX, E20
MOV SI, 2000H
MOV AX, [SI]
MOV BL, 10H
M DIV BL
OUT 80H, ALH
OUT 84H, DL ← ADD BP, DL
CALL DELAY
INC SI
RET

```

DISPLAY :

```

MOV AX, [SI]
MOV BL, 10H
DIV BL
ADD BP, AH
MOV AL, [BP]
OUT 80H, AL
OUT 84H, DL
CALL DELAY
INC SI
RET

```

DELAY :

```

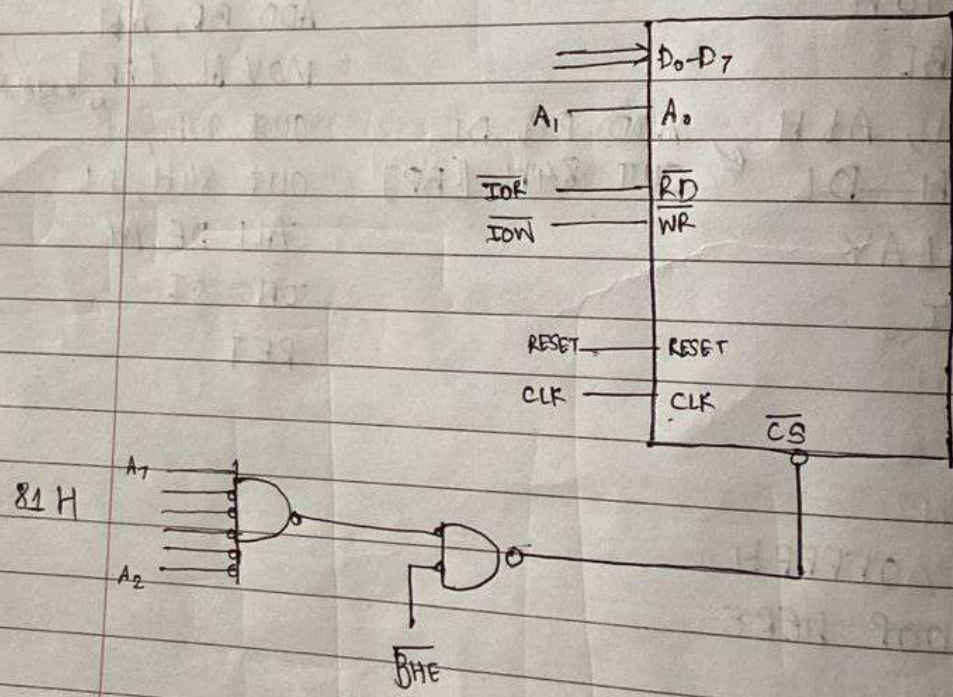
MOV CX, 0FFFFH
HERE : LOOP HERE
RET

```


MOV AL, 80H
OUT 86H, AL

REPEAT:
MOV BX, ES:2000H
MOV SI, 2000H
BACK:
MOV AL, [SI]
ADD

To interface 8279 to 8086 at base address 81H.



A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
1	0	0	0	0	0	0	1	
1	0	0	0	0	0	1	1	81H

max 64 lines can be interfaced
max 16 displays

62 — command port address
80 — data port address

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Write a program to display contents of memory location 3000 onwards in ~~write~~ ^{right} entry mode on 4 7 segment displays.

000 0000 11H 100

MOV AL, 11H

OUT 82H, AL

MOV AL, 34H

OUT 82H, AL

MOV AL,

OUT 82H, AL

MOV CX, 4

~~DELAY~~

DISPLAY:

MOV BX, 2000H

MOV AL, [SI]

INC SI

ADD BL, AL

MOV AL, [BX]

OUT 80H, AL

LOOP DISPLAY

Interface 8 7 seg display & 64 keys to 8279.

Sensor - switch (single pole, single throw)

FTPD Ram - data is read & lost

Ctrl & Shift 3 Scan 3 Return

Sensor RAM - bit RAM, sensor connected to bit closed - then closed, sensors for 64 locations.

All PC keyboards are strobed keyboards.

Strobed keyboard mode - scan lines not used, only return lines control pin func as strobe pin.

Scanning is done by ext hardware

Strobed keyboard meaning it has scanning hardware on it.

2 key lock out - 2 pressed simultaneously,
then 1 that's lifted 1st gets selected 1st

Bus contention - 2 devices trying to put conflicting data on bus

8257

Data bus & D₀-D₇ is multiplexed with A₈-A₁₅

needs to be demultiplexed with other pin ADSTB (address strobe)
↑ like ALE in 8086

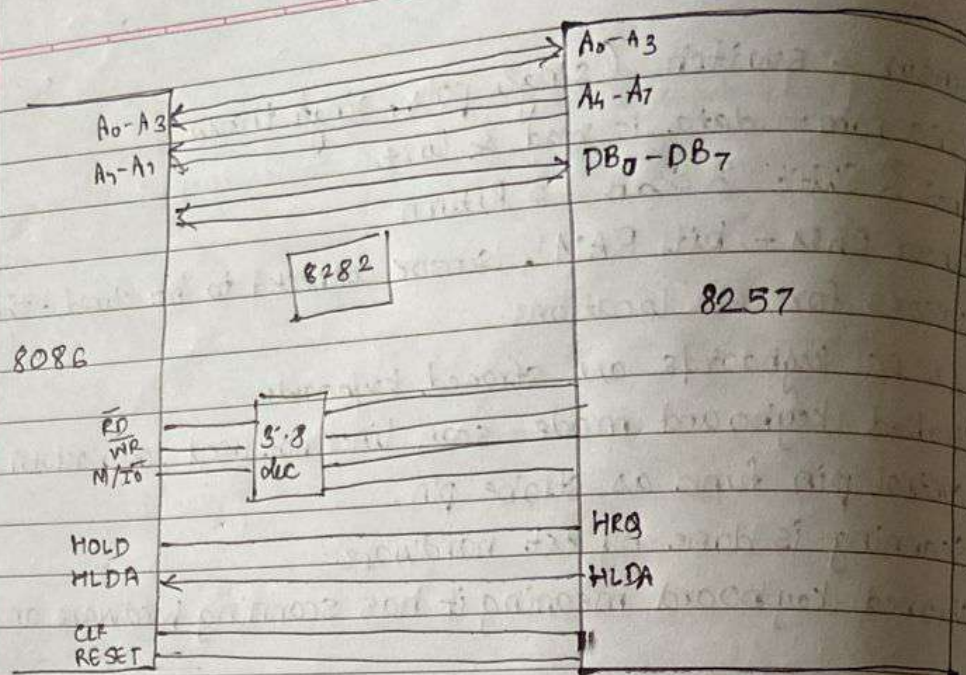
Interface 8257 to 8086 at starting address 80H. Show CP signals, show CPU side signal, 8 bit data bus, 16 bit address

To map

A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	0	0	0	0	0	0	0	80
1	0	0	1	1	1	1	0	9E

8081 numeric data processor

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Von Neumann Bottleneck

memory & i/o can't be accessed at the same time.

When memories were slow, we could insert wait state after T_2 (5 T states) using READY signal.

Memories are much faster, so not needed.

digit

$SL_0 - SL_3 =$ Scan lines, rows

$RL_0 - RL_7 =$ Column lines

Connecting $SL_0 - SL_2$ to 3:8 decoder we get 8 lines

encoded scan $SL_0 - SL_2$ connected to decoder, makes 1 0.

they get rotated

decoded scan - don't req external decoder

Internal operating freq of 8279 is 100 MHz

1st CW : 0 0 0 D D K K K

2nd CW : 0 0 1 P P P P P

eg: 8279 is being clocked 2 MHz, P P P P P ? $\frac{2 \times 10^6}{100 \times 10^3} = 20$

0 0 1 0 1 0 0
 3 4

10 100

MOV AL, 34H

OUT 82H, AL

Read FIFO/Sensor Ram

3rd CW : 0 1 0 A L X A A A

Interface 8279 for 16 keys & 4 displays with base address 80H.

(common cathode 7 seg displays)

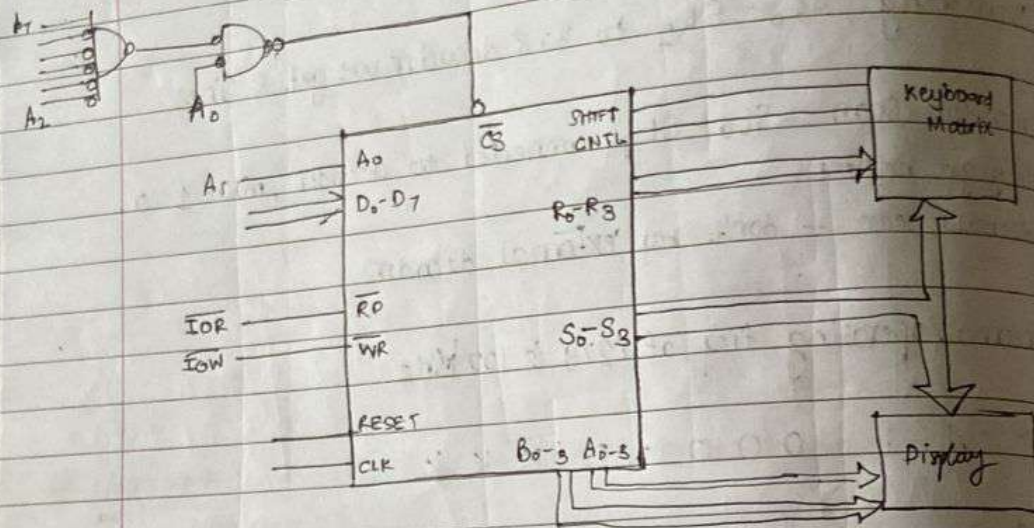
0 0 0 0 1 0 0 0 08H

0 0 1 1 0 1 0 0 34H

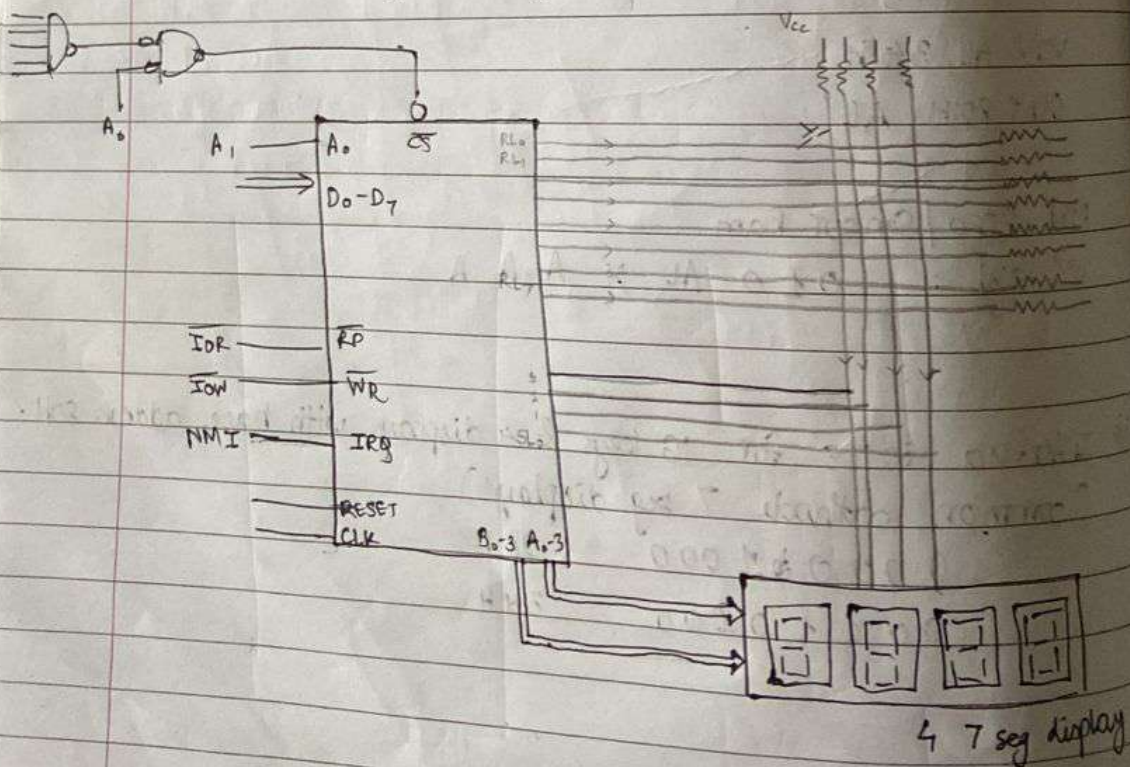
CS: 51 CALL 2000
CALL FAR

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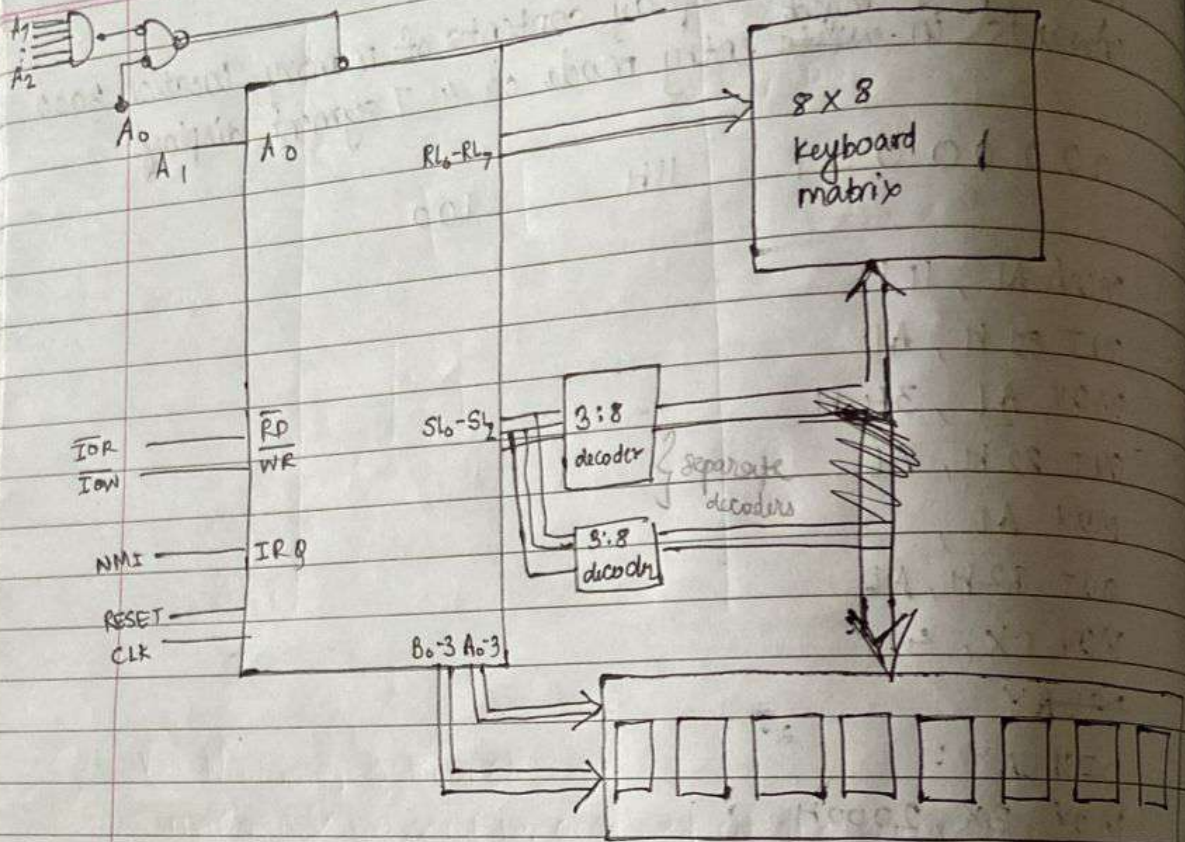
Interface 4 7seg displays & 32 keys to 8279.



In encoded form we cannot use SL_3 line for keyboard
 $\therefore$ no 4:16 dec

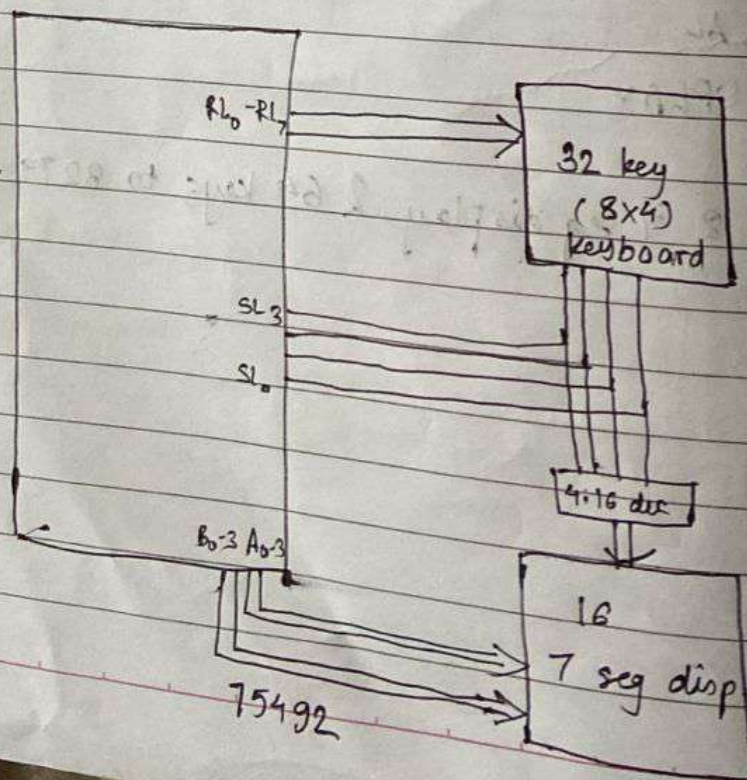
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8 7 seg display

Interface 16 7 seg displays & 32 keys



for each 4 lines
75491

boost current
carrying capacity